WATERWATCH · STEM CHECK

WaterWatch STEM check

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: PR-001 restated as ±30 % above 1.0 mg/L (O5) with the shared photometer as the monthly reference (O2), S31 within known science (E12: 8 of 8); S7 with PR-008 counting the hourly schedule only (O4); 36 within, 13 plausible across the options
0.1	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: first STEM check of the TRL 3 design (option A) and options B and C; battery capacity against its cells fixed (WWT-DEC-001 F1); independent review recorded, with fixes F5 to F8, nine new rows and S31 sent to Amish (O5)

WaterWatch STEM check

Verdict: the working principle of option A, an hourly timed flush through a small dark flow cell read by an 860 nm nephelometer and a membrane-free amperometric chlorine sensor, powered by a small panel and one LiFePO4 cell, sits within known science. One detail failed at the first check: the battery was specified as "1S2P 32700 cells, 3.2 V, 6 Ah", but one 32700 LiFePO4 cell already holds 6 Ah, so the stated capacity, the cell count and the thermal mass in the calculation disagreed. It was fixed under Amish's standing instruction of 2026-10-23 by specifying one 32700 cell (WWT-DEC-001 F1). The independent review then weakened 17 rows and found 7 missing claims; four more fixes were made under the same instruction (F5 to F8: a longer flush at low-pressure sites, a valve rated for 4 bar, TLS session resumption and a second check valve at the cell inlet), and its corrected chlorine error budget gave one row that failed with a fix that changes a requirement, so it went to Amish (S31; WWT-DEC-001 O5). Amish decided on 2026-10-25 ("i concur with your recommendations") to restate PR-001 as ±30 % above 1.0 mg/L (O5) and to make a shared DPD photometer the monthly reference (O2); with both, the budget meets all 8 cases and S31 is within known science (WWT-CAL-001 E11, E12). Of the 37 claims of option A, **28 are within known science and 9 are plausible, to confirm by test at TRL 4**. The plausible rows are the hot-day recharge (S4) and the sun shield factor it depends on (S5), the chlorine sensor's drift (S12), its pseudo-reference (S14), its flow dependence (S32) and low-conductivity water (S36), fouling (S22), freezing (S33) and the sun-heating of the temperature reading (S34). None can be closed by calculation, so `stem_check` is **open**. Options B and C have 6 claims each; each has 2 plausible rows and none fails. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports the calculation note's script `docs/04-calcs/sizing.py` (WWT-CAL-001 v0.6); the table is written by `docs/04-calcs/stm_table.py`.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`). S rows are option A, B rows option B, C rows option C.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	A 5 W panel covers the 0.54 Wh daily need on a dull rainy-season day	Physics	Electromagnetism and circuits	Energy balance: P_panel x peak sun hours x derating >= E_day; at 1 peak sun hour the yield is 3.0 Wh, still above the need	in 7.5 Wh/ day, out 0.5396 Wh/ day	Within known science	WWT-CAL-001 A3, B2
S2	The battery runs the unit for 7 days or more without sun	Physics	Electromagnetism and circuits	Autonomy = usable energy / daily need, with an 0.85 efficient 3.3 V converter (independent review): 15.36 Wh / 0.635 Wh; 28.5 days without the converter loss	24.2 days >= 7 days	Within known science	WWT-CAL-001 A6, B1
S3	The battery's cell count matches its stated capacity, 6 Ah at 3.2 V (19.2 Wh)	Chemistry	Electrochemistry and batteries	Pack capacity = cells in parallel x cell capacity; one 32700 LiFePO4 cell is 6.0 Ah (reseller listing; maker datasheet to obtain, WWT-DEC-001 to confirm 9); mismatch between stated and built capacity	0 % <= 5 %	Within known science	WWT-DEC-001 F1 (was 1S2P, 100 % mismatch)
S4	The panel refills an empty battery in 3 clear days or fewer, also on 45 degC days	Physics	Electromagnetism and circuits	Days = usable energy / (energy the charger may take - daily need); 1.19 days on a mild clear day; on a 45 degC day only 5.8 Wh may be charged under the shield (C4), so it depends on S5 (independent review)	2.94 days <= 3 days	Plausible, to confirm at TRL 3	Open
S5	Under the ventilated sun shield the enclosure stays near 51 degC on a 45 degC day and the battery can charge for about 5 h	Physics	Heat transfer	Lumped thermal model: absorbed sun (absorptance 0.70, 900 W/m2 direct) against 10 W/m2K over the exposed area, time constant 13 min; the shield passes a quarter of the solar gain (assumed)	Peak 50.8 degC, 5.4 sun hours below the 45 degC charge limit; the shield factor 0.25 is an assumption, to measure at TRL 4	Plausible, to confirm at TRL 3	Open
S6	One 90 s flush leaves less than 5 % of the previous hour's water in the cell	Physics	Fluid mechanics	Well-mixed cell: stale fraction = exp(-(Q t - V_tube) / V_cell), Q 0.5 L/ min, t 80 s, V_cell 0.185 L	2.9 % <= 5 %	Within known science	WWT-CAL-001 D3 (above 0.7 bar; see S29)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	when the chlorine is read						
S7	Hourly flushing uses 20 L/day or less at the regulator's upper tolerance	Physics	Fluid mechanics	Volume = 0.55 L/min x 90 s x 24 readings (pressure-compensating regulator, +10 %); ten confirming readings a month add about 0.25 L/day; PR-008 counts the hourly schedule only (WWT-DEC-001 O4, decided by Amish 2026-10-25)	19.8 L/day <= 20 L/day	Within known science	WWT-CAL-001 D5
S8	The open air-break vent at the cell outlet stops the drain hose siphoning the cell and stops drain water reaching the cell	Physics	Fluid mechanics	A siphon needs a sealed, liquid-filled path; a vent open to the atmosphere above the outlet holds atmospheric pressure there (hydrostatics). An air gap is the highest category of backflow protection in EN 1717 practice	The vent rises 60 mm above the outlet and stays open; the inlet side is S30	Within known science	WWT-PRC-001, How it works 1; WWT-DWG-001
S9	Bubbles of 100 um and larger clear the 66 mm cavity during the 30 s wait at 5 degC	Physics	Fluid mechanics	Stokes rise velocity, Re 0.24 (below 1, so Stokes holds), water at 5 degC; the wait clears bubbles of about 78 um and up, not bubbles stuck to the windows	18.4 s <= 30 s	Within known science	WWT-CAL-001 F1, F3
S10	An 860 nm LED with a 90 degree scatter detector and a 180 degree transmitted-light detector measures turbidity from 0 to 100 units	Physics	Optics and radiation	ISO 7027-1:2016 gives 860 nm diffuse radiation at 90 degrees for low turbidity (reported in FNU, about 0 to 40) and attenuation of the transmitted beam for higher turbidity (FAU); the cell has both detectors, so the upper range uses the 180 degree reading (independent review: say FNU, not NTU); Kelley et al. (2014) built an open-source turbidimeter on the same principle	Principle and wavelength match the standard; accuracy of this cell (R2) to verify with stabilised standards at TRL 4	Within known science	Kelley et al. 2014; ISO 7027-1
S11	At pH 8 and 25 degC about a	Chemistry		HOCl fraction = 1 / (1 + 10^(pH - pKa)), pKa from	0.004 absolute		

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	quarter of free chlorine is hypochlorous acid		Physical (kinetics, equilibria, thermochemistry)	Morris (1966): 3000 / T - 10.0686 + 0.0253 T	error vs 0.26 <= 0.02 absolute error vs 0.26	Within known science	WWT-CAL-001 E1, E2
S12	A membrane-free graphite three-electrode sensor reads free chlorine from 0 to 2 mg/L with drift below 0.15 mg/L per month	Chemistry	Electrochemistry and batteries	Amperometric reduction of HOCl at a carbon working electrode (Pan et al. 2015; pencil graphite electrodes, PLOS ONE 2021); long-term field drift of low-cost sensors is reported in Water Science and Technology (2025), not checked in full	Response within known science; the drift that R1 and R10 allow (about 0.15 mg/L per month with the shared photometer, WWT-CAL-001 E12; 0.13 before the decisions of 2026-10-25) is unverified	Plausible, to confirm at TRL 3	Open
S13	The chlorine error budget meets the PR-001 target at 0.5 mg/L and pH 7.5 before drift	Chemistry	Analytical	Root sum of squares of the DPD reference 0.07 mg/L, pH change between visits on the HOCl fraction and temperature residual, with the stale-water bias added linearly (independent review); the HOCl fraction is corrected for the measured temperature through pKa(T)	0.144 mg/L <= 0.2 mg/L	Within known science	WWT-CAL-001 E4, E9
S14	A silver/silver chloride pseudo-reference stays stable in low-chloride drinking water	Chemistry	Electrochemistry and batteries	The Ag/AgCl potential follows the chloride activity (Nernst); without a filled reference chamber it moves with the water's chloride	Potentiostat holds the working electrode against it; drift of the reference adds to S12; to measure at TRL 4	Plausible, to confirm at TRL 3	Open
S15	The latching valve opens at		Standards practice	A direct-acting solenoid valve needs no minimum	BOM line 13 specifies		WWT-CAL-001 D7;

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	every supply pressure from 0.5 to 4 bar	Engineering and technology		pressure difference (a pilot-operated valve needs about 0.2 to 0.5 bar), but a small direct-acting valve has a maximum opening differential (independent review)	direct acting and rated to open against at least 4 bar; to confirm in the datasheet	Within known science	WWT-DEC-001 F6
S16	The 48.3 x 3.2 mm pole carries a 35 m/s gust on the panel, sun shield and pole with a factor of at least 1.5 on yield	Physics	Strength of materials	sigma = M / Z at the footing, M 281 N m, Z 4.80 cm3; S235 yield 235 MPa / 1.5	59 MPa <= 157 MPa	Within known science	WWT-CAL-001 I2
S17	The 300 mm x 600 mm footing resists the gust load with a factor of at least 1.5, also in saturated sand	Physics	Mechanics and statics	Broms, short rigid free-head pile: H_u = 0.5 gamma D L^3 Kp / (e + L), Kp 3; dry medium sand (18 kN/m3) 890 N, saturated sand (9 kN/m3, independent review) shown; clay and made ground to assess at the survey visit	445 N >= 309 N	Within known science	WWT-CAL-001 I3, I4
S18	Charging is blocked below 0 degC and above 45 degC, which protects the LiFePO4 cell	Physics	Thermodynamics	Lithium plating on charge below 0 degC and faster ageing when charged hot are known for LiFePO4; cell datasheets give a 0 to 45 degC charge window	Charger with temperature cut-offs (BOM line 6) and NTC on the cell; to confirm in the charger datasheet (WWT-DEC-001, to confirm 8)	Within known science	WWT-DDR-001 D9
S19	One LTE-M upload, network attach included, costs about 20 mWh	Engineering and technology	Power and link budgets	SIM7000-class modem: about 150 mA at 3.8 V for the same 140 s per attach and post that S21 assumes (independent review)	claimed 20 mWh, estimate 22.2 mWh (ratio 0.90)	Within known science	WWT-CAL-001 A2
S20				Six sessions a day of 780 B payload; with TLS			WWT-DEC-001 F7;

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Uploads stay under 1 MB a month	Engineering and technology	Power and link budgets	session resumption about 1.2 kB overhead; a full handshake every session (about 6 kB) would give 1.22 MB (independent review), so the firmware keeps sessions	0.36 MB/ month <= 1 MB/month	Within known science	WWT-CAL-001 G1, G3
S21	An SMS leaves within 15 min of a confirmed crossing, with three tries	Engineering and technology	Control	Attach 120 s and send 20 s per try, 3 min between tries; the firmware bounds each try at 140 s; from the event itself, with retries, up to 92 min (H4)	13 min <= 15 min	Within known science	WWT-CAL-001 H1, H4
S22	A monthly wipe keeps the cell windows and the chlorine electrode clean enough	Biology	Microbiology	Biofilm and particle fouling of optical windows and electrodes in drinking water grows over weeks; residual chlorine slows it	Rate at a turbid site unknown; to observe at TRL 4 and in field trials	Plausible, to confirm at TRL 3	Open
S23	Alert thresholds of 0.2 mg/L free chlorine and 5 NTU match field guidance for the point of delivery	Biology	Epidemiology	WHO guidance as summarised for field use: at least 0.2 mg/L free chlorine at the point of delivery; chlorination loses effect as turbidity rises, markedly above about 10 NTU, so the 5 NTU alert is the earlier warning; the device reports measurements only (PR-013)	Defaults decided by Amish (WWT-DDR-001 D6); at 0.2 mg/L the budget is about +/-0.08 mg/ L, so a reading of 0.2 means about 0.12 to 0.28 (WWT-CAL-001 E3)	Within known science	WWT-PRB-001; WWT-DDR-001 D6
S24	Calibration chemicals are handled safely by a caretaker	Biology	Toxicology	DPD is harmful if swallowed and can sensitise skin, so tablets are handled with gloves and tested water goes to the drain, not back to a container; formazin is made from toxic precursors (hydrazine sulfate, hexamethylenetetramine),	Rule in WWT-PRC-001 Safety and the build plan; disposal with the operator	Within known science	WWT-PRC-001, Safety

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				so only prepared, stabilised standards are used			
S25	The enclosure (1.17 to 1.43 m) and the cell (0.81 to 0.90 m) sit within standing working reach; the panel is cleaned from the ground	Biology	Ergonomics	Comfortable standing work and reach heights for adults lie about 0.8 to 1.5 m (anthropometric tables, Pheasant and Haslegrave, Bodyspace); the panel at 2.16 m is out of reach for small adults (independent review), so it is wiped with a long-handled brush	Heights from the model; the 30 min monthly visit is a task analysis (WWT-CAL-001 J3)	Within known science	cad/src/model.py; WWT-CAL-001 J3
S26	The hot-day enclosure simulation uses a stable, converged time step	Mathematics	Numerical method	Explicit Euler is stable and within about 4 % of the exact first-order response for dt / tau below 0.1 (dt 60 s)	0.075 dt / tau <= 0.1 dt / tau	Within known science	WWT-CAL-001 C3, C4
S27	Flushed water (18 L/day at up to 2 mg/L free chlorine) to the tapstand basin is environmentally negligible	Chemistry	Environmental and atmospheric chemistry	Mass balance: 36 mg of chlorine a day (40 mg at the regulator tolerance), the same water the tap already delivers to the basin; option A adds no reagent, unlike option B, whose drain must go to a soakaway (B5)	Same water as the tap; no reagent added	Within known science	WWT-CAL-001 D2
S28	PVC, ASA, EPDM, polycarbonate and stainless parts withstand 0.2 to 2 mg/L free chlorine and outdoor sun	Chemistry	Materials and polymer chemistry	These materials are used in chlorinated water and outdoors at these levels; ASA and polycarbonate are UV-stabilised grades; EPDM is preferred over nitrile in chlorinated water	Wetted parts drain to waste behind a check valve, so taste and leaching do not reach users; grades to confirm	Within known science	WWT-PRC-001, Main components
S29	At a 0.5 bar site a 110 s flush leaves less than 5 % stale water	Physics	Fluid mechanics	As S6 at 0.35 L/min below the regulator's compensating pressure (90 s gives 8.5 %, independent review); 15.4 L/day	4.5 % <= 5 %	Within known science	WWT-DEC-001 F5; WWT-CAL-001 D8
S30							

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Cell water cannot flow back into the supply through the inlet when the main goes to negative pressure	Engineering and technology	Standards practice	On an intermittent supply the main can fall below atmospheric pressure and draw water back; one plain check valve is not a verifiable barrier (independent review). The cell is closed and full, so a back-siphon would pull cell water through the inlet	Two check valves in series (supply train and cell inlet) and the normally closed latching valve, shut 58 min in each hour (WWT-DEC-001 F8); a verifiable backflow device or an inlet air gap to assess with the operator under the local code (WWT-REG-001 R4)	Within known science	WWT-DEC-001 F8
S31	The chlorine error budget meets the PR-001 target at 1.5 mg/L at every pH	Chemistry	Analytical	Corrected budget as S13 (independent review): at 1.5 mg/L the pH terms dominate; a one-point calibration at 0.5 mg/L carries the reference error as a relative error at 1.5 mg/L. PR-001 restated by Amish on 2026-10-25 as +/-30 % above 1.0 mg/L (WWT-DEC-001 O5), with the shared DPD photometer (+/-0.03 mg/L +/-3 %) as the monthly reference (O2)	8 of 8 cases met calibrated at the reading level; worst FC 1.5 mg/L pH 8.5: +/-0.392 against +/-0.450 mg/L; with the photometer and calibrated at 0.5 mg/L where worse, 8 of 8 (WWT-CAL-001 E12; worst +/-0.411	Within known science	WWT-DEC-001 O2 and O5 (decided by Amish 2026-10-25); WWT-CAL-001 E11, E12

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					against +/-0.450 mg/L)		
S32	The membrane-free sensor's current does not depend on the flow past it	Chemistry	Electrochemistry and batteries	Amperometric currents depend on mass transport, so on flow; the pressure-compensating regulator holds the flow above 0.7 bar, but at 0.5 bar it is 30 % lower (independent review)	Flow dependence of this sensor unknown; read at the end of the flush at a held flow; to measure at TRL 4	Plausible, to confirm at TRL 3	Open
S33	The water-filled cell survives a 0 degC night	Physics	Heat transfer	At 0 degC air a small exposed volume can fall below freezing by night-sky radiation, and ice expands about 9 % (independent review)	Freezing margin not shown; at sites near 0 degC drain the cell for the cold season or insulate it; to assess with the partner	Plausible, to confirm at TRL 3	Open
S34	The temperature probe reads the water to +/-0.5 degC	Physics	Heat transfer	The DS18B20 is +/-0.5 degC; it is read after a flush of four cell volumes of supply water, but the black cell heats in the sun between flushes (independent review)	Sun-heating bias of the reading to measure at TRL 4	Plausible, to confirm at TRL 3	Open
S35	After 20 % capacity fade from hot-weather ageing the battery still runs 7 days without sun	Chemistry	Electrochemistry and batteries	LiFePO4 loses capacity faster when held above 45 degC (independent review); autonomy with 80 % of the capacity and the converter loss	19.4 days >= 7 days	Within known science	WWT-CAL-001 A6, B1
S36	The three-electrode sensor works in low-conductivity water	Chemistry	Electrochemistry and batteries	Below about 50 uS/cm the solution resistance between the electrodes grows and the potentiostat loses control (independent review)	Conductivity of candidate sources to survey; a conductivity floor to set at TRL 4	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S37	Closing the valve at 4 bar does not overstress the sample line	Physics	Fluid mechanics	Joukowsky surge dp = rho a v, tube velocity 0.57 m/s, wave speed about 350 m/s in small PE tube (estimate): 2.0 bar on top of 4 bar, against about 10 bar for 1/4 in PE tube (typical rating, BOM line 14)	6 bar <= 10 bar	Within known science	WWT-CAL-001 D9
B1	DPD colorimetry reads free chlorine from 0 to 2 mg/L in a 12 mm cuvette	Chemistry	Analytical	DPD reacts with free chlorine to a magenta colour read near 515 nm (ISO 7393-2; EPA 330.5 adaptations in commercial photometers)	Standard method; used by field photometers and online analysers	Within known science	WWT-OPT-001
B2	DPD and buffer reagents keep their strength for a month at up to 50 degC in the cabinet	Chemistry	Physical (kinetics, equilibria, thermochemistry)	Liquid DPD indicator oxidises and darkens with heat and air; suppliers give shelf lives at room temperature	Shelf life at 50 degC unknown; to assess with a supplier	Plausible, to confirm at TRL 3	Open
B3	80 mL of each reagent lasts 90 days at 0.03 mL per test and 24 tests a day	Mathematics	Units and dimensional analysis	Mass balance of reagent	in 80 mL, out 64.8 mL	Within known science	WWT-OPT-001
B4	The analyser needs about 0.8 Wh/day, within the 5 W panel	Engineering and technology	Power and link budgets	Option A's budget plus two 0.5 W pumps for 5 s and a 0.3 W stirrer for 30 s per test	claimed 0.8 Wh/day, estimate 0.68 Wh/day (ratio 1.18)	Within known science	Open
B5	Reacted sample with DPD and phosphate buffer can go to a soakaway	Biology	Toxicology	About 1.5 mL of reagent a day in 18 L of water; DPD (N,N-diethyl-p-phenylenediamine) is harmful if swallowed, so the drain must not reach the basin users fill from	Drain to a soakaway, not the basin; to assess with the operator	Plausible, to confirm at TRL 3	Open
B6	The same 860 nm LED and 90 degree detector read turbidity in the cuvette before the reagent goes in	Physics	Optics and radiation	Nephelometry as S10	Same principle as option A	Within known science	WWT-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
C1	ORP and pH electrodes estimate free chlorine within the R1 target	Chemistry	Electrochemistry and batteries	ORP follows the log of the HOCl activity and the pH, and other redox couples move it; it is used as a control signal, and tracks chlorine only loosely in the 0.2 to 2 mg/L range	Principle known; meeting +/-0.2 mg/L unlikely without frequent calibration	Plausible, to confirm at TRL 3	Open
C2	A 12 mm probe in a compression gland holds against 4 bar	Physics	Mechanics and statics	Push-out force = p x pi d^2 / 4 at 4 bar; small PVC compression glands hold well over 100 N (to confirm in the gland datasheet)	45.2 N <= 100 N	Within known science	WWT-OPT-001
C3	Probes in the riser read the water people draw, without flushing	Physics	Fluid mechanics	While the tap runs, the riser carries the supply water; between draws the water stands in the pipe and the reading is of stale water	Readings valid only during draws (a paddle flow switch wakes the unit); no reading at night when nobody draws	Within known science	WWT-OPT-001
C4	Insertion turbidity in the running pipe is not upset by bubbles and daylight	Physics	Optics and radiation	Entrained air from a tap opening scatters light like particles; the opaque manifold keeps daylight out	Bubble bias in flowing water unknown; to assess	Plausible, to confirm at TRL 3	Open
C5	With no valve and no flush the manifold needs about 0.4 Wh/ day	Engineering and technology	Power and link budgets	Uploads, sleep and status light as option A, plus 24 wake-ups of 20 s at 0.264 W	claimed 0.4 Wh/day, estimate 0.242 Wh/ day (ratio 1.65)	Within known science	Open
C6	Cutting the riser and fitting two unions keeps the tapstand watertight and serviceable	Engineering and technology	Standards practice	Threaded unions on 25 mm galvanised pipe are standard plumbing; it changes the tapstand, which the problem statement rules out	Within known science; conflicts with the PRB constraint of one tee only	Within known science	WWT-OPT-001

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. The kit's wording "to confirm at TRL 3" is kept in the table; for this repo, which is already at TRL 3, the plausible rows are confirmed by the tests planned for TRL 4 (WWT-PLN-001).

Requirements these rows support: S8, S28, S30 and S37 support PR-014 (supply protection); S3, S18 and S35 support PR-016 (battery handling); S24 supports PR-016 (calibration chemicals); the published calculation scripts support PR-015 (open files).

Fixes made under the standing instruction

Item	What failed	Fix (WWT-DEC-001)	Result
Battery capacity (S3)	BOM line 5, the precis and the calculation note specified "1S2P 32700 cells, 3.2 V, 6 Ah (19.2 Wh)". A 32700 LiFePO4 cell holds about 6 Ah (datasheet: 6000 mAh, 32 x 70 mm, 135 g), so two in parallel hold 12 Ah. The calculation used 6 Ah for energy but two cells' mass (0.29 kg) for heat. A builder ordering to the BOM would get a pack twice the stated capacity, or wrong cells	F1: one 32700 LiFePO4 cell (1S1P), 6 Ah, 19.2 Wh, 135 g, with its protection board, NTC and fuse in the same tray. Weighed against two 32700 cells (12 Ah, more autonomy, about USD 4 more, a parallel joint to weld) and two 26650 cells (6.4 Ah, two joints); one cell is the simplest to build and keeps every energy figure	Mismatch 0 % (was 100 %); thermal mass 1,305 J/K (was 1,460); enclosure peaks unchanged to 0.1 °C; autonomy 28.5 days; BOM price held at USD 20

| Low-pressure flush (S29, after the review of S6) | At 0.5 bar, below the regulator's 0.7 bar compensating pressure, the flow falls to about 0.35 L/min and a 90 s flush leaves 8.5 % stale water against 5 % | F5: a 110 s flush at sites below 0.7 bar, set per site at commissioning (a firmware setting, no hardware change). Weighed against a longer flush everywhere (wastes water at normal pressure) and a lower-threshold regulator (not a catalogue part at 0.5 L/min) | 4.5 % stale; 15.4 L/day; 35 mWh/day more awake time (WWT-CAL-001 D8) | | Valve rating (S15) | A small direct-acting valve has a maximum opening differential; the spec named only "no minimum" | F6: BOM line 13 asks for a valve rated to open against at least 4 bar | To confirm in the datasheet (WWT-DEC-001, to confirm) | | Data volume (S20) | With a full TLS handshake every session (about 6 kB) uploads come to 1.22 MB a month against 1 MB | F7: the firmware keeps TLS sessions (session resumption), about 1.2 kB overhead | 0.36 MB a month (WWT-CAL-001 G1, G3) | | Inlet backflow (S30, after the review of S8) | On an intermittent main the supply can go below atmospheric pressure; the inlet side then relied on one plain check valve, which is not a verifiable barrier | F8: a second check valve, 1/4 in push-fit inline, at the cell inlet, so two check valves and the normally closed latching valve stand between the cell and the supply. Weighed against an inlet air gap (needs a gravity-fed cell, a change of layout) and a verifiable backflow preventer (larger, dearer; to assess with the operator under the local code) | BOM line 14 +USD 3 (indicative); parts USD 325 |

The fixes do not change the working principle, the pitch or the scope. They are recorded in WWT-DEC-001 as decided under Amish's standing instruction of 2026-10-23, and he can reverse them. One row was not fixed under the standing instruction because its fix changes a requirement: S31, the chlorine error budget at 1.5 mg/L (6 of 8 cases met by the corrected budget, the worst missing by 0.017 mg/L; 1 of 4 when calibrated at 0.5 mg/L). Restating PR-001 as ±30 % above 1.0 mg/L meets all 8 cases (WWT-CAL-001 E11); fitting the pH probe at more sites or the shared photometer (WWT-DDR-004 D2) helps but does not close it alone. Amish decided O5 and O2 on 2026-10-25 as recommended: with the photometer as the monthly reference and the restated target, 8 of 8 cases are met, the worst at ±0.411 against ±0.450 mg/L (WWT-CAL-001 E12), so S31 is now within known science.

Fields not relevant

None. Every field is touched: physics (energy, heat transfer, fluid mechanics, optics, structure), chemistry (chlorine equilibria, electrochemistry, battery, materials, environment), biology (fouling, epidemiology of thresholds, toxicology of calibration chemicals, ergonomics), mathematics (units, numerical method) and engineering (link budgets, control, standards practice).

Concept options

- **Option A, timed-flush flow cell on its own pole:** the S rows above. Within known science; open on S5, S12, S14 and S22.
- **Option B, reagent colorimetric analyser cabinet:** within known science in principle (DPD colorimetry is the reference method, B1); reagent life in a hot cabinet (B2) and the discharge of reacted sample (B5) are plausible, to assess.
- **Option C, in-riser probe manifold:** within known science as plumbing and electrochemistry; ORP as a free chlorine estimate (C1) is plausible but unlikely to meet PR-001, and bubble bias in running water (C4) is to assess. Cutting the riser conflicts with the problem statement's constraint of one tee (C6), a scope point, not a science failure.

Independent review

The independent reviewer was a separate Claude session, started on 2026-10-25 with read access only to this document, the calculation note WWT-CAL-001, the product requirements WWT-PRS-001 and the sizing script with its printed output, and none of the author's reasoning. It was asked to refute the headline results and every "Within known science" verdict, and recomputed most numbers by hand. It upheld S2, S3, S9, S11, S16, S26 and the arithmetic of S5, S12, S14 and S22, refuted none outright, weakened 17 rows and listed 7 missing claims. What it found and what changed:

Row	Finding	Change
S1	"Overcast" was a rainy-season average of 2.5 peak sun hours	Relabelled; at 1 peak sun hour the panel still gives 3.0 Wh against 0.54 Wh
S2	No converter loss from the 2.5 to 3.65 V cell to the 3.3 V rail	0.85 efficiency added (WWT-CAL-001 A6): 24.2 days, still within; S3 renamed, the cell source is a reseller listing (maker datasheet to confirm)
S4	The 1.19-day refill assumes charging is never blocked by heat; on a 45 °C day only 5.8 Wh may be charged	S4 now uses the hot-day case: 2.94 days against 3, plausible and tied to S5
S5	Arithmetic reproduced (13.4 min time constant, 22.6 W absorbed); the shield gap also warms the air; "10.8 x" should read 10.7 x	Kept plausible; the shield factor and the heat transfer coefficient go to the TRL 4 temperature log; figure corrected in WWT-CAL-001
S6	Holds above 0.7 bar; at 0.5 bar the stale fraction is 8.5 %	Fixed: F5, a 110 s flush below 0.7 bar (S29, 4.5 %); a washout test at TRL 4
S7	19.8 L against 20 L leaves 1 %; ten confirming readings a month reach 20.1 L	Kept within for the hourly schedule; how PR-008 counts event flushes went to Amish (WWT-DEC-001 O4); decided 2026-10-25: the hourly schedule only
S8	The outlet vent holds, but the inlet side relied on one check valve on an intermittent main	Fixed: F8, a second check valve at the cell inlet (S30); a verifiable device or air gap to assess under the local code
S9	Upheld; the wait clears bubbles of about 78 µm and up, not bubbles stuck on windows	Note added; S9 does not support PR-002's accuracy
S10	ISO 7027 reports FNU; the 90 degree method covers about 0 to 40	Restated with the 180 degree transmitted reading for the upper range, in FNU
S13, S31	The reference error should scale with a one-point calibration; the stale bias should add linearly; temperature between visits and flow are missing	Stale bias added linearly and the temperature correction through pKa(T) stated (S13 still within at 0.5 mg/L, ±0.144 mg/L); new S31 failed at 1.5 mg/L with a fix that changes PR-001 (to Amish, O5; decided 2026-10-25, now within); new S32 for flow
S15	A direct-acting valve has a maximum opening differential	Fixed: F6, valve rated to open against 4 bar
S16	The sun shield enlarges the wind area by about 50 %	Wind area now the shield's: 59 MPa, factor 4.0 (WWT-CAL-001 I1, I2)
S17	Holds in dry sand; in saturated sand the capacity halves	Saturated case now governs: 445 N, factor 2.2 (I4); clay and made ground at the survey visit
S19	S21 assumes a 140 s attach; at 140 s the upload costs about 22 mWh	Same attach time in both rows: estimate 22 mWh, ratio 0.9
S20	A full TLS handshake every session gives 1.22 MB a month	Fixed: F7, TLS session resumption, 0.36 MB
S21	The 77 min event-to-SMS figure counted one try	Worst case with all three tries 92 min (H4); the 15 min limit on PR-005 is from the confirmed crossing and stands

Row	Finding	Change
S23	"Safe delivery" breaks PR-013's rule; 10 NTU against the 5 NTU threshold	Wording fixed; 5 NTU stated as the earlier warning; resolution at 0.2 mg/L stated (about ±0.08 mg/L)
S24	DPD called low hazard while B5 calls it harmful	Aligned: gloves, tested water to drain, disposal with the operator
S25	The panel at 2.16 m is out of reach for small adults	Panel wiped with a long-handled brush from the ground
S27	36 mg is the nominal; 40 mg at tolerance; drain destinations of A and B differ	Both figures given; A drains to the basin, B to a soakaway
Missing	Freezing; sun-heated temperature reading; battery ageing; low conductivity; water hammer; condensation, lightning and theft; evidence for the shield gap	Added S33 (plausible), S34 (plausible), S35 (within, 19.3 days after 20 % fade), S36 (plausible), S37 (within, 6.0 bar against about 10 bar); condensation, lightning and theft to the risk register (WWT-RSK-001); the shield gap stays under S5

The reviewer would keep S12, S14 and S22 plausible, as they are. Its answer is summarised row by row above; every change was made by the author after reading it.

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- EN 1717 (backflow protection; air gaps), as listed in WWT-REG-001 R4 (S8).